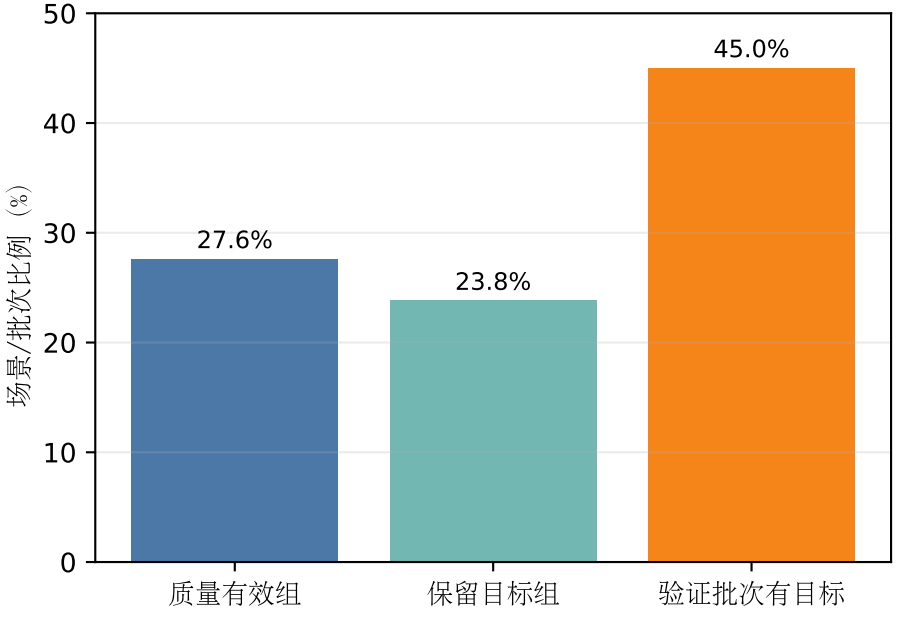
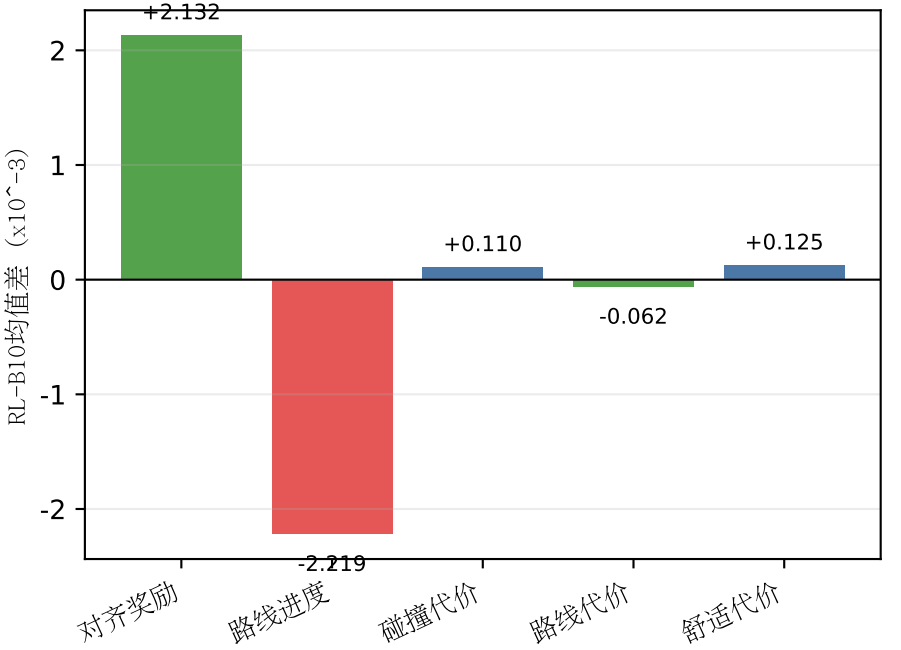


（a）训练集目标覆盖与实际更新信号



（b）validation-1000×3配对变化